Prashant Bhandari

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Summary

Electronics Engineering student specializing in PCB design, embedded firmware development, and IoT/robotics systems. Proven track record with ESP32, KiCad, C/C++, and MicroPython; shipped hardware from schematic through production across satellite, industrial, and consumer IoT domains. International competition finalist with hands-on leadership of 350+ students in embedded systems.

Technical Skills

PCB Design: KiCad, Proteus, Schematic Capture, PCB Layout, Buck/Boost Converter Design, DRC/ERC
Embedded Systems: ESP32, STM32 Bare-Metal C, NuttX RTOS, FreeRTOS, Arduino, ECU2000, HCM111Z, BLE, LoRa, UART/SPI/I2C
Programming: C/C++, Python, MicroPython, JavaScript
Hardware: IoT Systems, Power Electronics, QC Test Fixtures, Robotics, CubeSat Subsystems
Tools: Git, Fusion 360, 3D CAD, OLED Dashboards, Sensor Integration, OTA, JLCPCB/Gerber Workflows

Experience

Electronics Design Intern Sept 2025 – Jan 2026
SEDS Nepal (Remote) *Pokhara, Nepal*

- Designed and explored electronics for pseudo-satellite focusing on practical and modular subsystem architectures.
- Worked through component selection, schematic development, BOM preparation, and design reviews while considering reliability, power requirements, and system-level integration.
- Collaborated with team members from different engineering areas to understand how individual electronics modules fit into larger aerospace systems.

Electronics Engineer Intern Jan 2025 – Jul 2025
Yarsa Tech *Pokhara, Nepal*

- Designed, tested, and iterated buck converter circuits, improving PCB layouts and power regulation through multiple KiCad design revisions.
- Built an automated quality-control testing device for power backup products, developing custom C/C++ firmware to automate product testing and verification.
- Developed a BLE-enabled QR/barcode scanning system, integrating the scanner hardware with custom embedded firmware for a complete working prototype.
- Worked with embedded displays and user interfaces using LVGL, gaining practical experience in developing firmware and hardware for commercial products.

Projects

Micromouse Autonomous Maze-Solving Robot | *Custom PCB, PID, QTR Sensors*

- Designed and fabricated a custom PCB, reducing the robot's board size by 40%; implemented PID motion control and a flood-fill maze-solving algorithm on an Arduino Nano.
- **1st Place** LOCUS 2025 (IOE Pulchowk), **2nd Place** Delta 5.0 (IOE Purwanchal), **5th Place** Techfest IIT Bombay.

Smart Helmet Safety System | *IoT, BLE, SpO2, Fall Detection*

- Prototyped a wearable safety system integrating pulse/SpO2 sensing, fall detection, and SOS alerts, with sensor data transmitted via BLE to a supervisor monitoring dashboard.

DC–DC Buck Converter | *KiCad, XL2013E1, Power Electronics*

- Designed a 12V-to-5V buck converter from schematic to PCB, iterating the layout to achieve ∼90% efficiency and reduced output ripple.

Solar Greenhouse Automation System | *ESP32, BLE, MOSFET, OLED*

- Built a solar-powered greenhouse monitoring and control system using ESP32 to monitor soil moisture, temperature, and humidity, with automated irrigation, BLE connectivity, and an OLED dashboard.

Pocket CubeSat Module (1.5U PocketQube) | *Stacked PCB, LoRa, NodeMCU*

- Developed a compact stacked-PCB architecture integrating onboard computing, power management, sensors, and LoRa communication in a 1.5U form factor with Li-Po solar charging.

LoRaNet Communication Module | *KiCad, ESP32, LoRa, C/C++*

- Designed the schematic and multi-layer PCB for a long-range wireless communication system, achieving ∼7 km line-of-sight range.
- Developed ESP32 firmware for reliable packet transmission and communication with LoRa transceivers.

Education

Pashchimanchal Campus (Tribhuvan University) Pokhara, Nepal
B.E. in Electronics, Communication & Information Engineering *2023 – Present (4th Year)*

Leadership

Electronics & Software Lead 2024 – 2026
Robotics Club, Pashchimanchal Campus

- Coached 350+ students in PCB design, firmware development, and robotics; launched an in-house PCB prototyping lab.
- Spearheaded competition robotics projects; organized workshops and technical training programs campus-wide, including a 15-day "Rainy Session" training 100+ students.

Achievements

Top 25 Nationwide – U.S.–Nepal Startup Weekend Challenge (Safety Modules), 2026
International Finalist – Techfest IIT Bombay, representing Nepal (Safety Modules), 2025
1st Place – LOCUS 2025 Micromouse, IOE Pulchowk Campus
Runner-Up – Delta Robothon, 2025
1st Runner-Up – Autonomous Boat Racing, Robotics Club WRC 2024
Winner – Battle For Speed / Line Follower Competition, Robotics Club 2023